



Developing Generative AI Solutions

Introduction

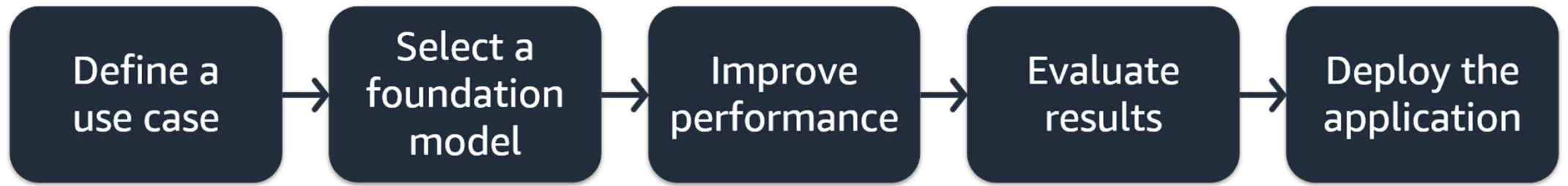
You will explore the generative artificial intelligence (generative AI) application lifecycle, which includes the following:

- Defining a business use case
- Selecting a foundation model (FM)
- Improving the performance of an FM
- Evaluating the performance of an FM
- Deployment and its impact on business objectives



Generative AI application lifecycle

This lifecycle encompasses several stages.



Defining a Use Case

Parts of a use case

- A well-defined business use case typically consists of the following parts:

Use case name

- A short and descriptive name that identifies the use case

Brief description

- A high-level summary of the use case's purpose and objective

Actors

- The entities or stakeholders that interact with the system or process
- These can be human actors (for example, customers or employees) or external systems.

Preconditions

- The conditions that must be true before the use case can be initiated

Basic flow (main success scenario)

- A step-by-step description of the actions and interactions that occur when the use case is completed successfully, from start to finish
- This is the primary path or happy path – for example, a list of each step necessary to achieve success.

Defining a Use Case

Alternative flows (extensions)

- Additional scenarios or paths that might occur due to exceptional conditions, errors, or alternative user choices
- These describe how the system should handle these situations – for example, contingency plans.

Postconditions

- The state or conditions that must be true after the successful completion of the use case

Business rules

- Any business policies, constraints, or regulations that govern the behavior of the system or process within the context of the use case

Nonfunctional requirements

- Any nonfunctional requirements, such as performance, security, or usability considerations, that are relevant to the use case

Assumptions

- Any assumptions made about the system, environment, or context that are necessary for the use case to be valid or applicable

Notes or additional information

- Any additional notes, explanations, or supplementary information that might be helpful for understanding or implementing the use case

Selecting an FM



Pre-trained model selection criteria

- Cost
- Modality
- Latency
- Multi-lingual support
- Model size
- Model complexity
- Customization
- Input/Output length
- Responsibility considerations
- Deployment and integration

A few pre-trained models on Amazon Bedrock



- **AI21 labs - Jurassic-2 (J2)** is AI21 Labs' state-of-the-art large language model (LLM). Supports cross-industry use cases including long-form and short-form text generation, contextual question answering, summarization, and classification.



- **Amazon Titan** foundation models are a family of models built by Amazon Web Services (AWS) that are pre-trained on large datasets, which makes them powerful, general-purpose models. There are three types of Amazon Titan models: embeddings, text generation, and image generation.

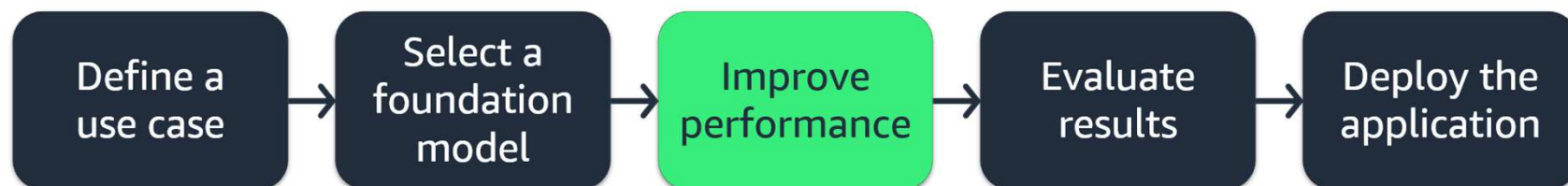


- **Meta - Llama** is a family of large language models that uses publicly available data for training. These models are based on the transformer architecture, which allows it to process input sequences of arbitrary length and generate output sequences of variable length.



- **Stability AI - Stable Diffusion** is an industry-leading image generation model. Stable Diffusion can generate images of from text input.

Improving the Performance of an FM



Prompt engineering is the fastest way to harness the power of large language models (LLMs)

Design: Crafting clear, unambiguous, and context-rich prompts that effectively communicate the desired task or output to the model

Augmentation: Incorporating additional information or constraints into the prompts, such as examples, demonstrations, or task-specific instructions, to guide the model's generation process

Tuning: Iteratively refining and adjusting the prompts based on the model's outputs and performance, often through human evaluation or automated metrics

Ensembling: Combining multiple prompts or generation strategies to improve the overall quality and robustness of the outputs

Mining: Exploring and identifying effective prompts through techniques like prompt searching, prompt generation, or prompt retrieval from large prompt libraries

Prompt Techniques

Prompt engineering techniques are strategies used to guide generative AI models. Some common prompt engineering techniques include the following:

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought (CoT) prompting
- Self-consistency
- Tree of thoughts (ToT)
- Retrieval Augmented Generation (RAG)
- Automatic Reasoning and Tool-use (ART)
- ReAct prompting

RAG

RAG is a natural language processing (NLP) technique that combines the capabilities of retrieval systems and generative language models to produce high-quality and informative text outputs.

RAG has several business applications:

- Building intelligent question-answering systems
- Expanding and enriching existing knowledge bases
- Generating high-quality content



Fine-tuning

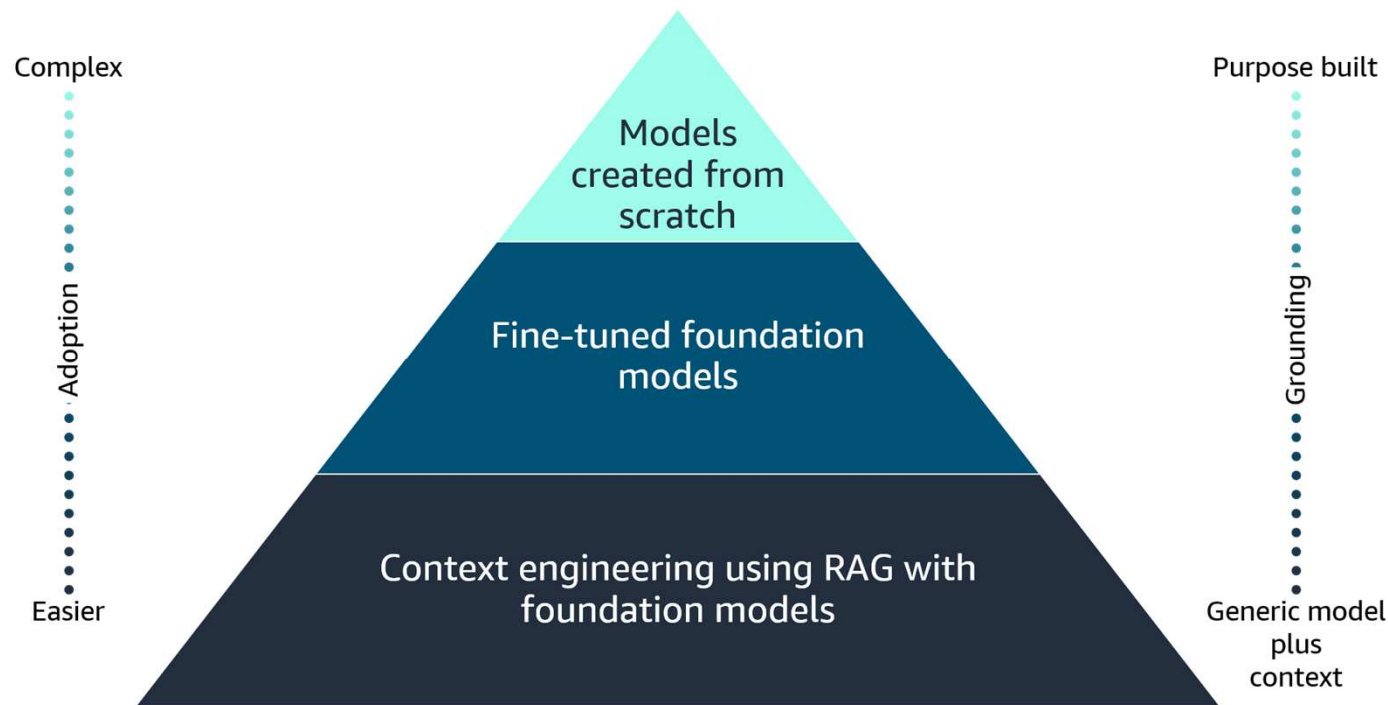
- Fine-tuning refers to the process of taking a pre-trained language model and further training it on a specific task or domain-specific dataset.
- Fine-tuning allows the model to adapt its knowledge and capabilities to better suit the requirements of the business use case.

2 ways to fine-tune a model

- Instruction fine-tuning
- Reinforcement learning from human feedback (RLHF)

Cost trade-offs of various approaches to foundation model customization

Trade-off between cost and accuracy when deciding whether to use a pre-trained foundation model or pursue a more customized approach.



Automated multi-step tasks with agents

- **Agents** are another component in generative AI solutions that can enhance the performance and capabilities of the foundation model.
- Agents play a crucial role in breaking down complex processes into smaller, manageable steps and orchestrating their completion.
- Agents are software components or entities designed to perform specific actions or tasks autonomously or semi-autonomously, based on predefined rules or algorithms.
- By using agents for multi-step tasks, organizations can achieve higher levels of automation, consistency, and efficiency

Examples of tasks that agents can accomplish:

- Task coordination
- Reporting and logging
- Scalability and concurrency
- Integration and communication

Evaluating Foundation Model Performance

Types of evaluation methods:

- **Human evaluation;** Human evaluators can provide qualitative feedback on factors like coherence, relevance, factuality, and overall quality of the model's outputs.
- **Benchmark datasets;** Benchmark datasets are curated collections of data designed specifically for evaluating the performance of language models or other AI systems. These datasets often consist of carefully selected examples or tasks that cover a wide range of topics, complexities, and linguistic phenomena.
- **Automated metrics;** Although human evaluation is considered the gold standard, automated metrics can provide a quick and scalable way to evaluate foundation model performance. These metrics typically measure specific aspects of the model's outputs, such as the following:
 - Perplexity (a measure of how well the model predicts the next token)
 - BLEU score (for evaluating machine translation)
 - F1 score (for evaluating classification or entity recognition tasks)

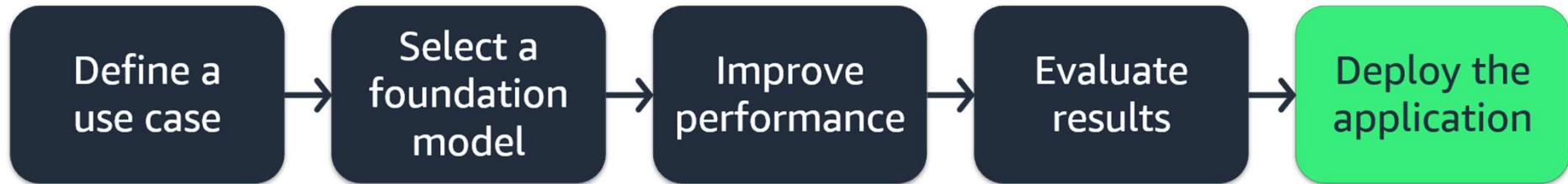
Relevant Metrics

Metrics like ROUGE, BLEU, and BERTScore provide an initial assessment of the foundation model's capabilities.

- **Recall-Oriented Understudy for Gisting Evaluation (ROUGE)** is a set of metrics used for evaluating automatic summarization and machine translation systems. It measures the quality of a generated summary or translation by comparing it to one or more reference summaries or translations.
- **Bilingual Evaluation Understudy (BLEU)** evaluates the quality of machine-generated text, particularly in the context of machine translation. It measures the similarity between a generated text and one or more reference translations, considering both precision and brevity.
- **BERTScore** evaluates the semantic similarity between a generated text and one or more reference texts. Uses pre-trained Bidirectional Encoder Representations from Transformers (BERT) models to compute contextualized embeddings for the input texts, and then calculates the cosine similarity between them.

Deploying the Application

Key factors that you should consider when deploying your model on premises or in the cloud.



- **Cost:** Pay for the resources that you use with no minimum fees.
- **Regions:** Model deployment is limited to certain AWS Regions.
- **Quotas:** Ensure that you have the adequate service resources for your AWS account.
- **Security:**
 - If your model is deployed in AWS infrastructure, the security responsibility is shared between the company and AWS.
 - If accessing a model outside of AWS, security considerations must be evaluated for data leaving the AWS account.

Knowledge Check

Q1.

A developer is creating a real-time translation application for mobile devices.

Which criterion would be most important when selecting a pre-trained model for this task?

- A. Model size
- B. Model complexity
- C. Latency
- D. Customization

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Q2.

You are working on developing an intelligent question-answering system for a company's internal knowledge base. The system must provide accurate and relevant answers to employees' queries by using the extensive information available in the knowledge base.

Which business application of Retrieval Augmented Generation (RAG) would be the most suitable in this scenario?

- A. Generating high-quality content, such as articles, reports, or summaries
- B. Building intelligent question-answering systems
- C. Expanding and enriching existing knowledge bases
- D. Training a large language model on the knowledge base

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Q3.

You are developing a large-scale data processing pipeline that involves multiple steps, such as data ingestion, cleaning, transformation, and analysis. Each step requires different computational resources and dependencies.

What would be the most important role of agents to ensure efficient and reliable completion of this multi-step task?

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Q4.

You are developing a state-of-the-art natural language processing (NLP) model for text summarization.

Which evaluation method would be most appropriate to ensure the model's performance and quality?

- A. Automated testing
- B. Using benchmark datasets
- C. Code review
- D. Human evaluation

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Q5.

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- B. BERTScore
- C. Recall-Oriented Understudy for Gisting Evaluation (ROUGE)
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